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JEE MAIN · QUESTION PAPER

JEE Main 2025 (Online) 28th January Morning Shift

JEE Main Question Paper with Answer Key

28 Jan 2025	75	English
HELD ON	QUESTIONS	LANGUAGE

INSIDE THIS PDF

Chemistry	25 questions
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Answer key	at the end

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

The incorrect decreasing order of atomic radii is

(A) $\mathrm{Si} > \mathrm{P} > \mathrm{Cl} > \mathrm{F}$

(B) $\mathrm{Mg} > \mathrm{Al} > \mathrm{C} > \mathrm{O}$

(C) $\mathrm{Al} > \mathrm{B} > \mathrm{N} > \mathrm{F}$

(D) $\mathrm{Be} > \mathrm{Mg} > \mathrm{Al} > \mathrm{Si}$

Open solution

Q2. MCQ (Single Correct Answer)

What is the freezing point depression constant of a solvent, 50 g of which contain 1 g non volatile solute (molar mass $256\ \mathrm{g\ mol^{-1}}$) and the decrease in freezing point is 0.40 K ?

(A) $4.43\ \mathrm{K\ kg\ mol^{-1}}$

(B) $3.72\ \mathrm{K\ kg\ mol^{-1}}$

(C) $5.12\ \mathrm{K\ kg\ mol^{-1}}$

(D) $1.86\ \mathrm{K\ kg\ mol^{-1}}$

Open solution

Q3. MCQ (Single Correct Answer)

Given below are two statements:

Statement I : D-glucose pentaacetate reacts with 2,4 -dinitrophenylhydrazine

Statement II : Starch, on heating with concentrated sulfuric acid at $100^{\circ}\mathrm{C}$ and $2-3\mathrm{atmosphere}$ pressure produces glucose.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Open solution

Q4. MCQ (Single Correct Answer)

A weak acid HA has degree of dissociation x . Which option gives the correct expression of (pH - $\mathrm{pK_a}$)?

(A) $\log\left(\frac{1-x}{x}\right)$

(B) 0

(C) $\log(1+2x)$

(D) $\log\left(\frac{x}{1-x}\right)$

Open solution

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Q5. MCQ (Single Correct Answer)

Open solution

$[A]_0$ / mol L^{-1}	$t_{1/2}$ / min
0.100	200
0.025	100

For a given reaction $R \rightarrow P$, $t_{1/2}$ is related to $[A]_0$ as given in table.

Given: $\log 2 = 0.30$

Which of the following is true?

A. The order of the reaction is $1/2$.

B. If $[A]_0$ is 1 M , then $t_{1/2}$ is $200\sqrt{10}$ min

C. The order of the reaction changes to 1 if the concentration of reactant changes from 0.100 M to 0.500 M.

D. $t_{1/2}$ is 800 min for $[A]_0 = 1.6M$

Choose the correct answer from the options given below:

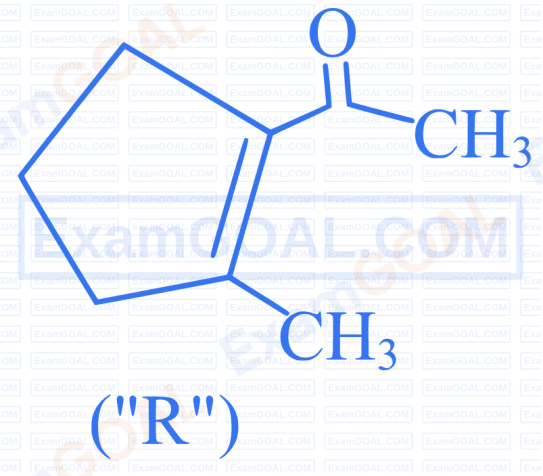
(A) A and C Only **(B)** A, B and D Only **(C)** A and B Only **(D)** C and D Only

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Q6. MCQ (More than One Correct Answer)

Open solution

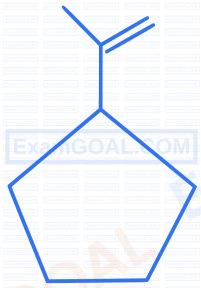
A molecule ("P") on treatment with acid undergoes rearrangement and gives ("Q"). ("Q") on ozonolysis followed by reflux under alkaline condition gives (" R "). The structure of (" R ") is given below.



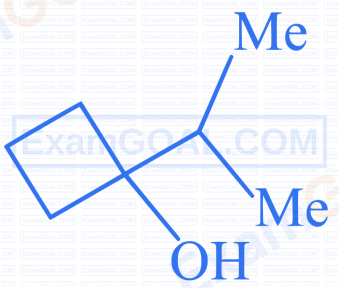
("R")

The structure of ("P") is

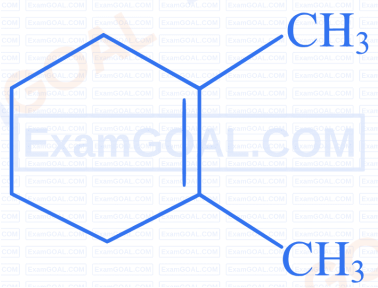
(A)



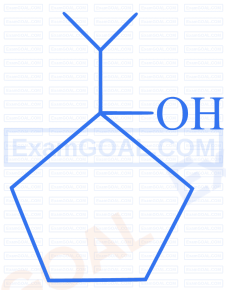
(B)



(C)



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(D) 

Q7. MCQ (Single Correct Answer)

Open solution

The products A and B in the following reactions, respectively are

$$A \xleftarrow{Ag-NO_2} CH_3-CH_2-CH_2-Br \xrightarrow{AgCN} B$$

(A) $CH_3-CH_2-CH_2-ONO, CH_3-CH_2-CH_2-CN$

(B) $CH_3-CH_2-CH_2-NO_2, CH_3-CH_2-CH_2-CN$

(C) $CH_3-CH_2-CH_2-ONO, CH_3-CH_2-CH_2-NC$

(D) $CH_3-CH_2-CH_2-NO_2, CH_3-CH_2-CH_2-NC$

Q8. MCQ (Single Correct Answer)

Open solution

In a multielectron atom, which of the following orbitals described by three quantum numbers will have same energy in absence of electric and magnetic fields?

A. $n=1, l=0, m_l=0$

B. $n=2, l=0, m_l=0$

C. $n=2, l=1, m_l=1$

D. $n=3, l=2, m_l=1$

E. $n=3, l=2, m_l=0$

Choose the correct answer from the options given below:

(A) B and C Only **(B)** C and D Only **(C)** A and B Only **(D)** D and E Only

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Q9. MCQ (Single Correct Answer)

Open solution

Given below are two statements:

Statement I: In the oxalic acid vs $KMnO_4$ (in the presence of dil H_2SO_4) titration the solution needs to be heated initially to $60^\circ C$, but no heating is required in Ferrous ammonium sulphate (FAS) vs $KMnO_4$ titration (in the presence of dil H_2SO_4)

Statement II: In oxalic acid vs $KMnO_4$ titration, the initial formation of $MnSO_4$ takes place at high temperature, which then acts as catalyst for further reaction. In the case of FAS vs $KMnO_4$, heating oxidizes Fe^{2+} into Fe^{3+} by oxygen of air and error may be introduced in the experiment.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q10. MCQ (Single Correct Answer)

Open solution

Consider the following elements In, Tl, Al, Pb, Sn and Ge . The most stable oxidation states of elements with highest and lowest first ionisation enthalpies, respectively, are

(A) +4 and +1 (B) +4 and +3 (C) +1 and +4 (D) +2 and +3

Q11. MCQ (Single Correct Answer)

Open solution

Both acetaldehyde and acetone (individually) undergo which of the following reactions?

A. Iodoform Reaction

B. Cannizzaro Reaction

C. Aldol Condensation

D. Tollen's Test

E. Clemmensen Reduction

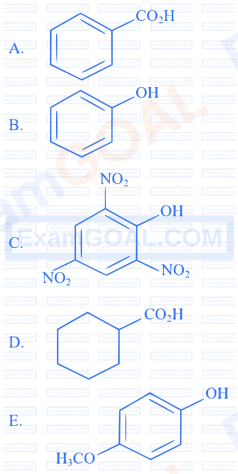
Choose the correct answer from the options given below:

(A) C and E Only (B) A, B and D Only (C) A, C and E Only (D) B, C and D Only

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Q12. MCQ (Single Correct Answer)

The compounds that produce CO_2 with aqueous $NaHCO_3$ solution are:



Open solution

Choose the correct answer from the options given below:

(A) A and C Only **(B)** A, B and E Only **(C)** A and B Only **(D)** A, C and D Only

Q13. MCQ (Single Correct Answer)

The metal ion whose electronic configuration is not affected by the nature of the ligand and which gives a violet colour in non-luminous flame under hot condition in borax bead test is

(A) Cr^{3+} **(B)** Ti^{3+} **(C)** Ni^{2+} **(D)** Mn^{2+}

Open solution

Q14. MCQ (Single Correct Answer)

Consider ' n ' is the number of lone pair of electrons present in the equatorial position of the most stable structure of ClF_3. The ions from the following with ' n ' number of unpaired electrons are

A. V^{3+}

B. Ti^{3+}

C. Cu^{2+}

D. Ni^{2+}

E. Ti^{2+}

Choose the correct answer from the options given below:

(A) B and D Only **(B)** A, D and E Only **(C)** B and C Only **(D)** A and C Only

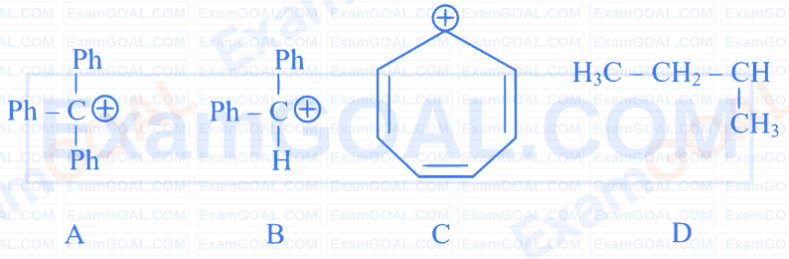
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Q15. MCQ (Single Correct Answer)

Open solution

The correct order of stability of following carbocations is :



(A) A > B > C > D (B) C > B > A > D (C) B > C > A > D (D) C > A > B > D

Q16. MCQ (Single Correct Answer)

Open solution

Match the List - I with List - II

	List - I (Redox Reaction)		List - II (Type of Redox Reaction)
(A)	$CH_{4(g)} + 2O_{2(g)} \xrightarrow{\Delta} CO_{2(g)} + 2H_2O_{(l)}$	(I)	Disproportionation reaction
(B)	$2NaH_{(s)} \xrightarrow{\Delta} 2Na_{(s)} + H_{2(g)}$	(II)	Combination reaction
(C)	$V_2O_{5(s)} + 5Ca_{(s)} \xrightarrow{\Delta} 2\,V_{(s)} + 5CaO_{(s)}$	(III)	Decomposition reaction
(D)	$2H_2O_{2(aq)} \xrightarrow{\Delta} 2H_2O_{(l)} + O_{2(g)}$	(IV)	Displacement reaction

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II (B) A-IV, B-I, C-II, D-III

(C) A-II, B-III, C-I, D-IV (D) A-II, B-III, C-IV, D-I

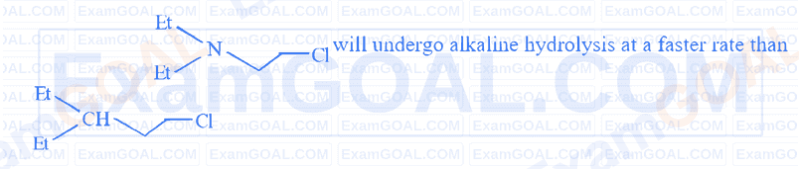
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Q17. MCQ (Single Correct Answer)

Open solution

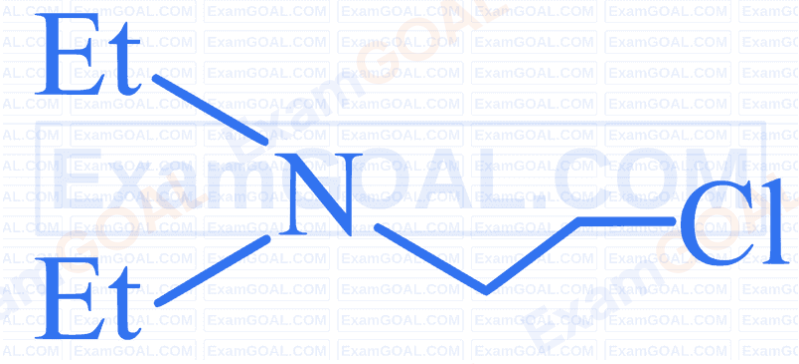
Given below are two statements :

Statement I :



will undergo alkaline hydrolysis at a faster rate than

Statement II : In



, intramolecular substitution takes place first by involving lone pair of electrons on nitrogen.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Both Statement I and Statement II are correct

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

Q18. MCQ (Single Correct Answer)

Open solution

Which of the following oxidation reactions are carried out by both $K_2Cr_2O_7$ and $KMnO_4$ in acidic medium?

A. $I^- \rightarrow I_2$

B. $S^{2-} \rightarrow S$

C. $Fe^{2+} \rightarrow Fe^{3+}$

D. $I^- \rightarrow IO_3^-$

E. $S_2O_3^{2-} \rightarrow SO_4^{2-}$

Choose the correct answer from the options given below :

(A) A, B and C Only **(B)** C, D and E Only **(C)** B, C and D Only **(D)** A, D and E Only

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Q19. MCQ (Single Correct Answer)

The molecules having square pyramidal geometry are

(A) BrF_5 & PCl_5

(B) SbF_5 & PCl_5

(C) BrF_5 & $XeOF_4$

(D) SbF_5 & $XeOF_4$

Open solution

Q20. MCQ (Single Correct Answer)

Ice and water are placed in a closed container at a pressure of 1 atm and temperature 273.15 K . If pressure of the system is increased 2 times, keeping temperature constant, then identify correct observation from following

(A) Liquid phase disappears completely.

(B) The amount of ice decreases.

(C) The solid phase (ice) disappears completely.

(D) Volume of system increases .

Open solution

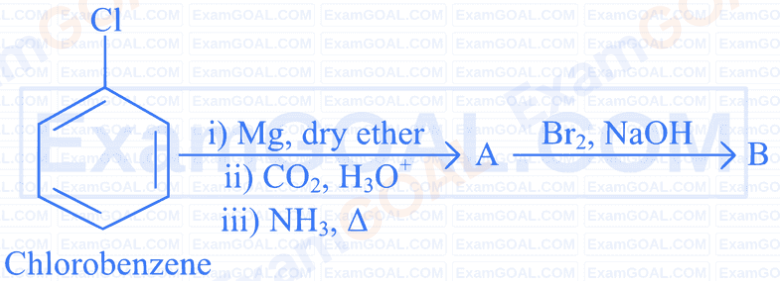
Q21. Numerical

The formation enthalpies, $\Delta H_f^{\ominus}$ for $H_{(g)}$ and $O_{(g)}$ are 220.0 and $250.0\ kJ\ mol^{-1}$, respectively, at 298.15 K , and $\Delta H_f^{\ominus}$ for $H_2O_{(g)}$ is $-242.0\ kJ\ mol^{-1}$ at the same temperature. The average bond enthalpy of the $O-H$ bond in water at 298.15 K is _____ $kJ\ mol^{-1}$ (nearest integer).

Open solution

Q22. Numerical

Consider the following sequence of reactions:



11.25 mg of chlorobenzene will produce _____ $\times 10^{-1}$ mg of product B.

(Consider the reactions result in complete conversion.)

[Given molar mass of C, H, O, N and Cl as 12, 1, 16, 14 and $35.5\ g\ mol^{-1}$ respectively]

Open solution

Q23. Numerical

The molarity of a 70% (mass/mass) aqueous solution of a monobasic acid (X) is _____ $\times 10^{-1}$ M (Nearest integer)

[Given: Density of aqueous solution of (X) is $1.25\ g\ mL^{-1}$

Molar mass of the acid is $70\ g\ mol^{-1}$]

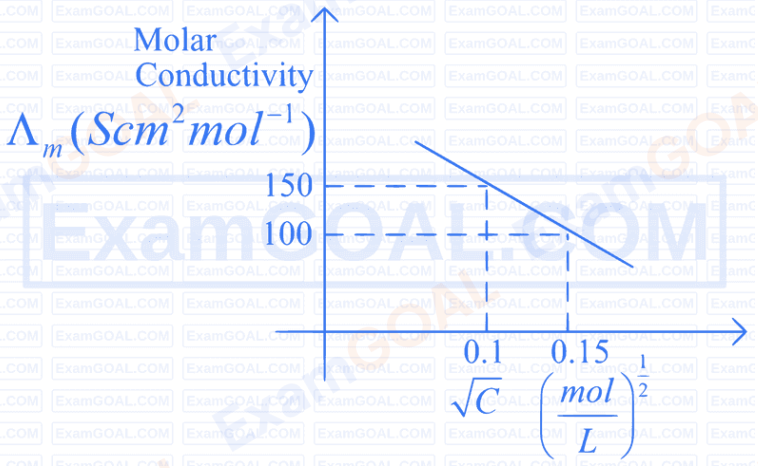
Open solution

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Q24. Numerical

Given below is the plot of the molar conductivity vs $\sqrt{\text{concentration}}$ for KCl in aqueous solution.

Open solution



If, for the higher concentration of KCl solution, the resistance of the conductivity cell is 100Ω, then the resistance of the same cell with the dilute solution is ' x ' Ω

The value of x is _____ (Nearest integer)

Q25. Numerical

Open solution

Quantitative analysis of an organic compound (X) shows following % composition.

C : 14.5%

Cl : 64.46%

H: 1.8 %

Empirical formula mass of the compound (X) is _____ $\times 10^{-1}$

(Given molar mass in $g\ mol^{-1}$ of $C : 12, H : 1, O : 16, Cl : 35.5$)

Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Open solution

Let $A(x, y, z)$ be a point in xy-plane, which is equidistant from three points $(0, 3, 2), (2, 0, 3)$ and $(0, 0, 1)$.

Let $B = (1, 4, -1)$ and $C = (2, 0, -2)$. Then among the statements

(S1) : $\triangle ABC$ is an isosceles right angled triangle, and

(S2) : the area of $\triangle ABC$ is $\frac{9\sqrt{2}}{2}$,

(A) both are false

(B) only (S2) is true

(C) only (S1) is true

(D) both are true

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Q27. MCQ (Single Correct Answer)

If $f(x)=\frac{2^x}{2^x+\sqrt{2}}, \mathrm{x} \in \mathbb{R}$, then $\sum_{\mathrm{k}=1}^{81} f\left(\frac{\mathrm{k}}{82}\right)$ is equal to

(A) 82 (B) $81\sqrt{2}$ (C) 41 (D) $\frac{81}{2}$

Open solution

Q28. MCQ (Single Correct Answer)

The sum, of the squares of all the roots of the equation $x^2+|2x-3|-4=0$, is

(A) $6(2-\sqrt{2})$ (B) $3(3-\sqrt{2})$ (C) $3(2-\sqrt{2})$ (D) $6(3-\sqrt{2})$

Open solution

Q29. MCQ (Single Correct Answer)

The relation $R=\{(x, y): x, y \in \mathbb{Z}$ and $x+y$ is even $\}$ is:

(A) reflexive and transitive but not symmetric

(B) reflexive and symmetric but not transitive

(C) an equivalence relation

(D) symmetric and transitive but not reflexive

Open solution

Q30. MCQ (Single Correct Answer)

Let $\langle a_{\mathrm{n}}\rangle$ be a sequence such that $a_0=0, a_1=\frac{1}{2}$ and $2a_{\mathrm{n}+2}=5a_{\mathrm{n}+1}-3a_{\mathrm{n}}, \mathrm{n}=0,1,2,3, \ldots$. Then $\sum_{k=1}^{100} a_k$ is equal to

(A) $3a_{100}+100$ (B) $3a_{100}-100$ (C) $3a_{99}-100$ (D) $3a_{99}+100$

Open solution

Q31. MCQ (Single Correct Answer)

$\cos\left(\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{5}{13}+\sin^{-1}\frac{33}{65}\right)$ is equal to:

(A) $\frac{33}{65}$ (B) 1 (C) $\frac{32}{65}$ (D) 0

Open solution

Q32. MCQ (Single Correct Answer)

If $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{96x^2\cos^2 x}{(1+e^x)}\mathrm{d}x=\pi\left(\alpha\pi^2+\beta\right), \alpha, \beta \in \mathbb{Z}$, then $(\alpha+\beta)^2$ equals

(A) 196 (B) 100 (C) 64 (D) 144

Open solution

Q33. MCQ (Single Correct Answer)

Let T_{r} be the rth term of an A.P. If for some m, $\mathrm{T}_{\mathrm{m}}=\frac{1}{25}$, $\mathrm{T}_{25}=\frac{1}{20}$, and $20\sum_{\mathrm{r}=1}^{25}\mathrm{T}_{\mathrm{r}}=13$, then $5\,\mathrm{m}\sum_{\mathrm{r}=\mathrm{m}}^{2\mathrm{m}}\mathrm{T}_{\mathrm{r}}$ is equal to

(A) 98 (B) 126 (C) 112 (D) 142

Open solution

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Q34. MCQ (Single Correct Answer)

Let ${}^nC_{r-1} = 28$, ${}^nC_r = 56$ and ${}^nC_{r+1} = 70$. Let $A(4\cos t, 4\sin t)$, $B(2\sin t, -2\cos t)$ and $C\left(3r - n, r^2 - n - 1\right)$ be the vertices of a triangle ABC, where t is a parameter. If $(3x-1)^2 + (3y)^2 = \alpha$, is the locus of the centroid of triangle ABC , then α equals

(A) 18 (B) 8 (C) 20 (D) 6

Open solution

Q35. MCQ (Single Correct Answer)

Two number k_1 and k_2 are randomly chosen from the set of natural numbers. Then, the probability that the value of $i^{k_1} + i^{k_2}$, $(i = \sqrt{-1})$ is non-zero, equals

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$

Open solution

Q36. MCQ (Single Correct Answer)

Three defective oranges are accidentally mixed with seven good ones and on looking at them, it is not possible to differentiate between them. Two oranges are drawn at random from the lot. If x denotes the number of defective oranges, then the variance of x is

(A) $26/75$ (B) $14/25$ (C) $18/25$ (D) $28/75$

Open solution

Q37. MCQ (Single Correct Answer)

Let ABCD be a trapezium whose vertices lie on the parabola $y^2 = 4x$. Let the sides AD and BC of the trapezium be parallel to y-axis. If the diagonal AC is of length $\frac{25}{4}$ and it passes through the point $(1, 0)$, then the area of $ABCD$ is

(A) $\frac{75}{8}$ (B) $\frac{125}{8}$ (C) $\frac{25}{2}$ (D) $\frac{75}{4}$

Open solution

Q38. MCQ (Single Correct Answer)

If the image of the point $(4, 4, 3)$ in the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-1}{3}$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to

(A) 12 (B) 9 (C) 7 (D) 8

Open solution

Q39. MCQ (Single Correct Answer)

Let for some function $y = f(x)$, $\int_0^x tf(t)dt = x^2 f(x)$, $x > 0$ and $f(2) = 3$. Then $f(6)$ is equal to

(A) 1 (B) 6 (C) 2 (D) 3

Open solution

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Q40. MCQ (Single Correct Answer)

The sum of all local minimum values of the function

$$f(x)=\begin{cases}1-2x, & x<-1\\ \frac{1}{3}(7+2|x|), & -1\le x\le 2\\ \frac{11}{18}(x-4)(x-5), & x>2\end{cases}$$

is

(A) $\frac{167}{72}$ (B) $\frac{157}{72}$ (C) $\frac{171}{72}$ (D) $\frac{131}{72}$

Open solution

Q41. MCQ (Single Correct Answer)

The number of different 5 digit numbers greater than 50000 that can be formed using the digits 0 , $1, 2, 3, 4, 5, 6, 7$, such that the sum of their first and last digits should not be more than 8 , is

(A) 5720 (B) 5719 (C) 4608 (D) 4607

Open solution

Q42. MCQ (Single Correct Answer)

Let the equation of the circle, which touches x-axis at the point $(a, 0), a>0$ and cuts off an intercept of length b on y-axis be $x^2+y^2-\alpha x+\beta y+\gamma=0$. If the circle lies below x-axis, then the ordered pair $\left(2a, b^2\right)$ is equal to

(A) $\left(\alpha, \beta^2+4\gamma\right)$ (B) $\left(\alpha, \beta^2-4\gamma\right)$

(C) $\left(\gamma, \beta^2-4\alpha\right)$ (D) $\left(\gamma, \beta^2+4\alpha\right)$

Open solution

Q43. MCQ (Single Correct Answer)

Let $f: \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x)=(2+3a)x^2+\left(\frac{a+2}{a-1}\right)x+b, a\neq 1$. If $f(x+y)=f(x)+f(\mathrm{y})+1-\frac{2}{7}x\mathrm{y}$, then the value of $28\sum_{i=1}^{5}|f(i)|$ is

(A) 735 (B) 675 (C) 715 (D) 545

Open solution

Q44. MCQ (Single Correct Answer)

Let O be the origin, the point A be $z_1=\sqrt{3}+2\sqrt{2}i$, the point $B\left(z_2\right)$ be such that $\sqrt{3}\left|z_2\right|=\left|z_1\right|$ and $\arg\left(z_2\right)=\arg\left(z_1\right)+\frac{\pi}{6}$. Then

(A) area of triangle ABO is $\frac{11}{4}$

(B) area of triangle ABO is $\frac{11}{\sqrt{3}}$

(C) ABO is a scalene triangle

(D) ABO is an obtuse angled isosceles triangle

Open solution

Q45. MCQ (Single Correct Answer)

The area (in sq. units) of the region $\left\{(x, \mathrm{y}): 0\le \mathrm{y}\le 2|x|+1, 0\le \mathrm{y}\le x^2+1, |x|\le 3\right\}$ is

(A) $\frac{32}{3}$ (B) $\frac{64}{3}$ (C) $\frac{17}{3}$ (D) $\frac{80}{3}$

Open solution

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Q46. Numerical

Let M denote the set of all real matrices of order 3×3 and let $S = \{-3, -2, -1, 1, 2\}$. Let

$S_1 = \{A = [a_{ij}] \in M : A = A^T \text{ and } a_{ij} \in S, \forall i, j\}$,

$S_2 = \{A = [a_{ij}] \in M : A = -A^T \text{ and } a_{ij} \in S, \forall i, j\}$,

$S_3 = \{A = [a_{ij}] \in M : a_{11} + a_{22} + a_{33} = 0 \text{ and } a_{ij} \in S, \forall i, j\}$.

If $n(S_1 \cup S_2 \cup S_3) = 125\alpha$, then α equals _____.

Open solution

Q47. Numerical

Let $f(x) = \begin{cases} 3x, & x < 0 \\ \min\{1 + x + [x], x + 2[x]\}, & 0 \le x \le 2 \\ 5, & x > 2 \end{cases}$

where [.] denotes greatest integer function. If α and β are the number of points, where f is not continuous and is not differentiable, respectively, then $\alpha + \beta$ equals _____ .

Open solution

Q48. Numerical

If $\alpha = 1 + \sum_{r=1}^{6} (-3)^{r-1} {}^{12}C_{2r-1}$, then the distance of the point $(12, \sqrt{3})$ from the line $\alpha x - \sqrt{3}y + 1 = 0$ is _____.

Open solution

Q49. Numerical

Let $E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1$ be an ellipse. Ellipses E_i 's are constructed such that their centres and eccentricities are same as that of E_1, and the length of minor axis of E_i is the length of major axis of $E_{i+1} (i \ge 1)$. If A_i is the area of the ellipse E_i, then $\frac{5}{\pi}\left(\sum_{i=1}^{\infty} A_i\right)$, is equal to _____.

Open solution

Q50. Numerical

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{d} = \vec{a} \times \vec{b}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|$, $|\vec{c} - 2\vec{a}|^2 = 8$ and the angle between $\vec{d}$ and $\vec{c}$ is $\frac{\pi}{4}$, then $|10 - 3\vec{b} \cdot \vec{c}| + |\vec{d} \times \vec{c}|^2$ is equal to _____.

Open solution

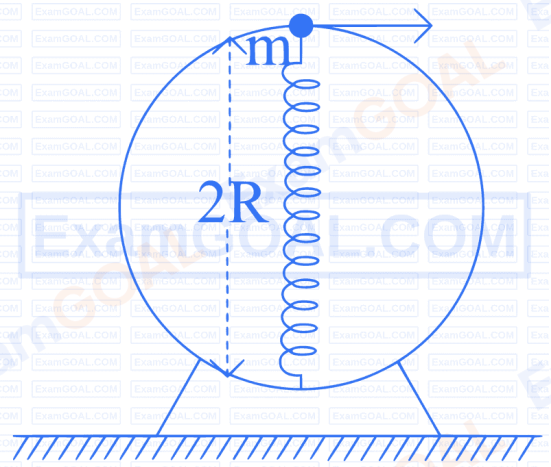
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Physics 25 questions

Q51. MCQ (Single Correct Answer)

Open solution

A bead of mass ' m ' slides without friction on the wall of a vertical circular hoop of radius ' R ' as shown in figure. The bead moves under the combined action of gravity and a massless spring (k) attached to the bottom of the hoop. The equilibrium length of the spring is ' R '. If the bead is released from top of the hoop with (negligible) zero initial speed, velocity of bead, when the length of spring becomes ' R ', would be (spring constant is ' k ', g is acceleration due to gravity)



(A) $\sqrt{2Rg+\frac{kR^2}{m}}$

(B) $\sqrt{3Rg+\frac{kR^2}{m}}$

(C) $\sqrt{2Rg+\frac{4kR^2}{m}}$

(D) $2\sqrt{gR+\frac{kR^2}{m}}$

Q52. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason $\mathbf{R}$

Assertion A: A sound wave has higher speed in solids than gases.

Reason R: Gases have higher value of Bulk modulus than solids.

In the light of the above statements, choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(C) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

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Q53. MCQ (Single Correct Answer)

Open solution

In the experiment for measurement of viscosity ' η ' of given liquid with a ball having radius R, consider following statements.

A. Graph between terminal velocity V and R will be a parabola.

B. The terminal velocities of different diameter balls are constant for a given liquid.

C. Measurement of terminal velocity is dependent on the temperature.

D. This experiment can be utilized to assess the density of a given liquid.

E. If balls are dropped with some initial speed, the value of η will change.

Choose the correct answer from the options given below:

(A) C, D and E Only (B) A, B and E Only (C) A, C and D Only (D) B, D and E Only

Q54. MCQ (Single Correct Answer)

Open solution

The center of mass of a thin rectangular plate (fig - x) with sides of length a and b, whose mass per unit area (σ) varies as $\sigma = \frac{\sigma_0 x}{ab}$ (where σ_0 is a constant), would be _____.

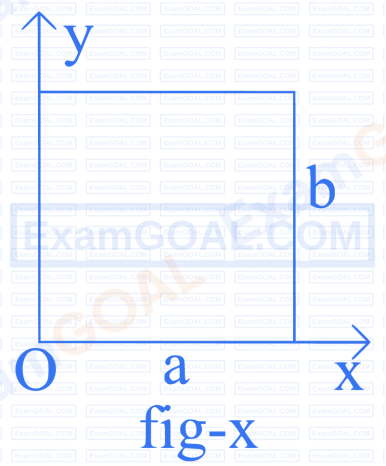


fig-x

(A) $\left(\frac{2}{3}a, \frac{2}{3}b\right)$ (B) $\left(\frac{2}{3}a, \frac{\mathrm{b}}{2}\right)$

(C) $\left(\frac{1}{3}a, \frac{\mathrm{b}}{2}\right)$ (D) $\left(\frac{a}{2}, \frac{\mathrm{b}}{2}\right)$

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Q55. MCQ (Single Correct Answer)

Open solution

Consider following statements:

A. Surface tension arises due to extra energy of the molecules at the interior as compared to the molecules at the surface, of a liquid.

B. As the temperature of liquid rises, the coefficient of viscosity increases.

C. As the temperature of gas increases, the coefficient of viscosity increases

D. The onset of turbulence is determined by Reynold's number.

E. In a steady flow two stream lines never intersect.

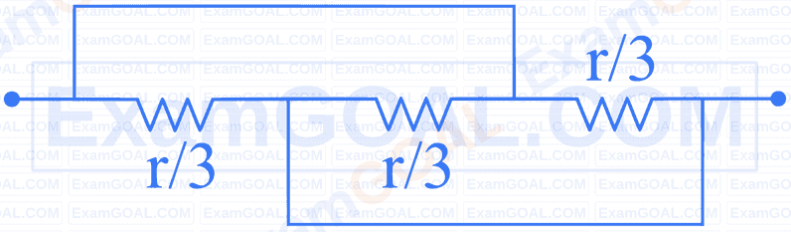
Choose the correct answer from the options given below:

(A) B, C, D Only (B) C, D, E Only (C) A, D, E Only (D) A, B, C Only

Q56. MCQ (Single Correct Answer)

Open solution

Find the equivalent resistance between two ends of the following circuit



(A) r (B) $\frac{r}{6}$ (C) $\frac{r}{3}$ (D) $\frac{r}{9}$

Q57. MCQ (Single Correct Answer)

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason $\mathbf{R}$

Assertion A: In a central force field, the work done is independent of the path chosen.

Reason R: Every force encountered in mechanics does not have an associated potential energy.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

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Q58. MCQ (Single Correct Answer)

A proton of mass ' m_P ' has same energy ' E ' as that of a photon of wavelength ' λ '. If the proton is moving at non-relativistic speed, then ratio of its de Broglie wavelength to the wavelength of photon is.

Open solution

(A) $\frac{1}{c}\sqrt{\frac{E}{m_p}}$

(B) $\frac{1}{\mathrm{c}}\sqrt{\frac{2\mathrm{E}}{\mathrm{m_p}}}$

(C) $\frac{1}{2\mathrm{c}}\sqrt{\frac{\mathrm{E}}{\mathrm{m_p}}}$

(D) $\frac{1}{\mathrm{c}}\sqrt{\frac{\mathrm{E}}{2\,\mathrm{m_p}}}$

Q59. MCQ (Single Correct Answer)

A wire of resistance R is bent into an equilateral triangle and an identical wire is bent into a square. The ratio of resistance between the two end points of an edge of the triangle to that of the square is

Open solution

(A) $8/9$

(B) $9/8$

(C) $32/27$

(D) $27/32$

Q60. MCQ (Single Correct Answer)

Consider a long thin conducting wire carrying a uniform current I. A particle having mass " m " and charge " q " is released at a distance " a " from the wire with a speed v_0 along the direction of current in the wire. The particle gets attracted to the wire due to magnetic force. The particle turns round when it is at distance x from the wire. The value of x is [μ_0 is vacuum permeability]

Open solution

(A) $a\left[1-\frac{mv_o}{2q\mu_0\mathrm{I}}\right]$

(B) $ae^{-\frac{4\pi\mathrm{mv_o}}{q\mu_o\mathrm{I}}}$

(C) $a\left[1-\frac{\mathrm{mv}}{\mathrm{q}}\mu_\mathrm{o}\mathrm{I}\right]$

(D) $\frac{a}{2}$

Q61. MCQ (Single Correct Answer)

Due to presence of an em-wave whose electric component is given by $E=100\sin(\omega t-kx)\mathrm{NC}^{-1}$ a cylinder of length 200 cm holds certain amount of em-energy inside it. If another cylinder of same length but half diameter than previous one holds same amount of em-energy, the magnitude of the electric field of the corresponding em-wave should be modified as

Open solution

(A) $50\sin(\omega\mathrm{t}-\mathrm{kx})\mathrm{NC}^{-1}$

(B) $400\sin(\omega\mathrm{t}-\mathrm{kx})\mathrm{NC}^{-1}$

(C) $200\sin(\omega t-kx)\mathrm{NC}^{-1}$

(D) $25\sin(\omega\mathrm{t}-\mathrm{kx})\mathrm{NC}^{-1}$

Q62. MCQ (Single Correct Answer)

Three infinitely long wires with linear charge density λ are placed along the $x-axis, y-axis$ and $z-$ axis respectively. Which of the following denotes an equipotential surface?

Open solution

(A) $\left(x^2+y^2\right)\left(y^2+z^2\right)\left(z^2+x^2\right)=$ constant

(B) $xyz=$ constant

(C) $xy+yz+zx=$ constant

(D) $(x+y)(y+z)(z+x)=$ constant

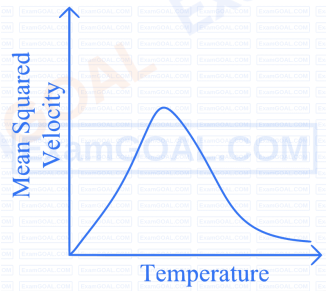
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Q63. MCQ (Single Correct Answer)

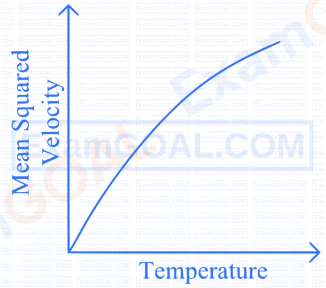
Open solution

For a particular ideal gas which of the following graphs represents the variation of mean squared velocity of the gas molecules with temperature?

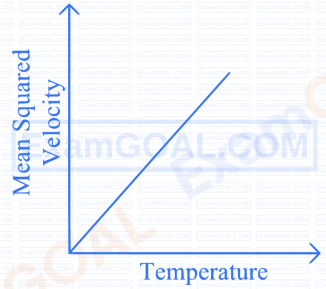
(A)



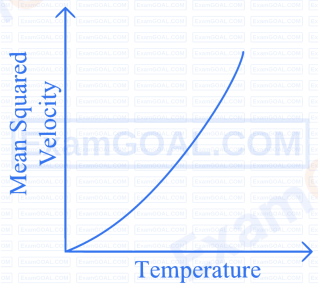
(B)



(C)



(D)

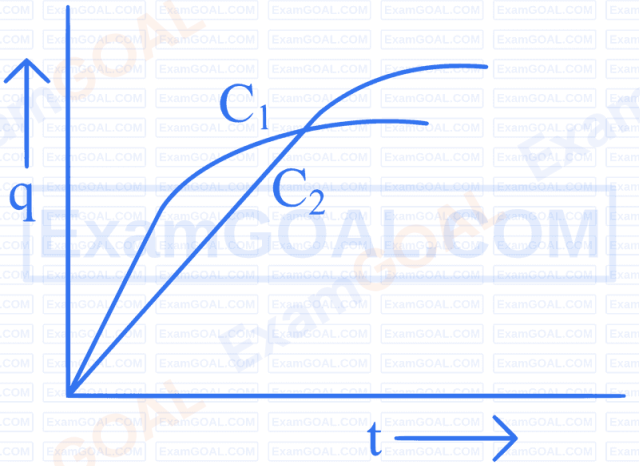


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Q64. MCQ (Single Correct Answer)

Two capacitors C_1 and C_2 are connected in parallel to a battery. Charge-time graph is shown below for the two capacitors. The energy stored with them are U_1 and U_2, respectively. Which of the given statements is true?

Open solution



(A) $C_2 > C_1, U_2 > U_1$

(B) $C_1 > C_2, U_1 > U_2$

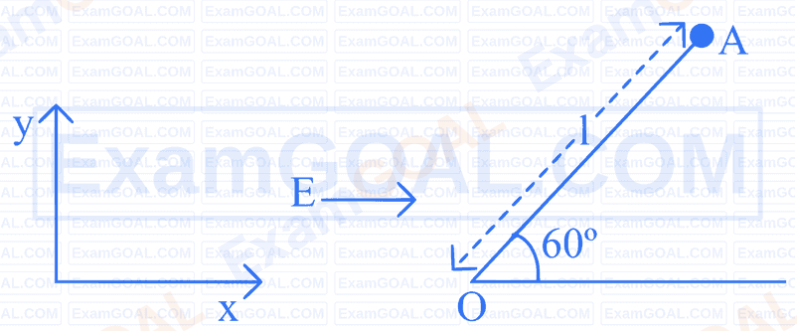
(C) $C_2 > C_1, U_2 < U_1$

(D) $C_1 > C_2, U_1 < U_2$

Q65. MCQ (Single Correct Answer)

A particle of mass ' m ' and charge ' q ' is fastened to one end ' A ' of a massless string having equilibrium length l, whose other end is fixed at point ' O '. The whole system is placed on a frictionless horizontal plane and is initially at rest. If uniform electric field is switched on along the direction as shown in figure, then the speed of the particle when it crosses the x-axis is

Open solution



(A) $\sqrt{\frac{qEl}{2m}}$

(B) $\sqrt{\frac{qEl}{4m}}$

(C) $\sqrt{\frac{qEl}{m}}$

(D) $\sqrt{\frac{2qEl}{m}}$

Q66. MCQ (Single Correct Answer)

A thin prism P_1 with angle 4° made of glass having refractive index 1.54 , is combined with another thin prism P_2 made of glass having refractive index 1.72 to get dispersion without deviation. The angle of the prism P_2 in degrees is

Open solution

(A) 1.5

(B) 16/3

(C) 3

(D) 4

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Q67. MCQ (Single Correct Answer)

Open solution

Which of the following circuits has the same output as that of the given circuit?



(A)



(B)



(C)



(D)

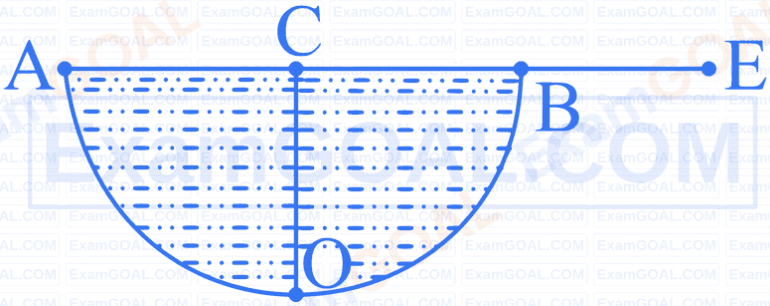


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Q68. MCQ (Single Correct Answer)

A hemispherical vessel is completely filled with a liquid of refractive index μ. A small coin is kept at the lowest point (O) of the vessel as shown in figure. The minimum value of the refractive index of the liquid so that a person can see the coin from point E (at the level of the vessel) is _____.

Open solution



(A) $\frac{3}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) $\sqrt{2}$

Q69. MCQ (Single Correct Answer)

Choose the correct nuclear process from the below options [p : proton, n : neutron, e^{-}: electron, e^{+}: positron, v : neutrino, $\bar{v}$: antineutrino]

Open solution

(A) $n \rightarrow p+e^{+}+v$ (B) $\mathrm{n} \rightarrow \mathrm{p}+\mathrm{e}^{-}+\bar{v}$

(C) $\mathrm{n} \rightarrow \mathrm{p}+\mathrm{e}^{+}+\bar{v}$ (D) $\mathrm{n} \rightarrow \mathrm{p}+\mathrm{e}^{-}+v$

Q70. MCQ (Single Correct Answer)

A Carnot engine (E) is working between two temperatures 473 K and 273 K . In a new system two engines - engine E_1 works between 473 K to 373 K and engine E_2 works between 373 K to 273 K . If η_{12}, η_1 and η_2 are the efficiencies of the engines E, E_1 and E_2, respectively, then

Open solution

(A) $\eta_{12}=\eta_1 \eta_2$ (B) $\eta_{12}=\eta_1+\eta_2$ (C) $\eta_{12} \geq \eta_1+\eta_2$ (D) $\eta_{12}<\eta_1+\eta_2$

Q71. Numerical

Two iron solid discs of negligible thickness have radii R_1 and R_2 and moment of inertia I_1 and I_2, respectively. For $R_2=2R_1$, the ratio of I_1 and I_2 would be $1/x$, where $\mathrm{x}=$ _____ .

Open solution

Q72. Numerical

In a measurement, it is asked to find modulus of elasticity per unit torque applied on the system. The measured quantity has dimension of $\left[M^a L^b T^c\right]$. If $b=3$, the value of c is _____.

Open solution

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Q73. Numerical

The moment of inertia of a solid disc rotating along its diameter is 2.5 times higher than the moment of inertia of a ring rotating in similar way. The moment of inertia of a solid sphere which has same radius as the disc and rotating in similar way, is n times higher than the moment of inertia of the given ring. Here, $\mathrm{n} =$ _____ Consider all the bodies have equal masses.

Open solution

Q74. Numerical

A tiny metallic rectangular sheet has length and breadth of 5 mm and 2.5 mm , respectively. Using a specially designed screw gauge which has pitch of 0.75 mm and 15 divisions in the circular scale, you are asked to find the area of the sheet. In this measurement, the maximum fractional error will be $\frac{x}{100}$ where x is _____ .

Open solution

Q75. Numerical

A double slit interference experiment performed with a light of wavelength 600 nm forms an interference fringe pattern on a screen with 10 th bright fringe having its centre at a distance of 10 mm from the central maximum. Distance of the centre of the same 10 th bright fringe from the central maximum when the source of light is replaced by another source of wavelength 660 nm would be _____ mm .

Open solution

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Answer Key

Chemistry

1. **D**	2. **C**	3. **C**	4. **D**	5. **B**
6. **A, D**	7. **D**	8. **D**	9. **C**	10. **B**
11. **C**	12. **D**	13. **C**	14. **B**	15. **D**
16. **D**	17. **A**	18. **A**	19. **C**	20. **C**
21. **466**	22. **93**	23. **125**	24. **150**	25. **1655**

Mathematics

26. **C**	27. **D**	28. **A**	29. **C**	30. **B**
31. **D**	32. **B**	33. **B**	34. **C**	35. **A**
36. **D**	37. **D**	38. **B**	39. **A**	40. **B**
41. **D**	42. **B**	43. **B**	44. **D**	45. **B**
46. **1613**	47. **5**	48. **5**	49. **54**	50. **6**

Physics

51. **B**	52. **C**	53. **C**	54. **B**	55. **B**
56. **D**	57. **B**	58. **D**	59. **C**	60. **B**
61. **C**	62. **A**	63. **C**	64. **A**	65. **C**
66. **C**	67. **A**	68. **D**	69. **B**	70. **D**
71. **16**	72. **0**	73. **4**	74. **3**	75. **11**

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